Dr. Qin Wang

[Homepage] | [LinkedIn] | [Github]

Sydney, Australia E-mail: qin.wang@data61.csiro.au

PORTFOLIO

- A **Research Scientist** (equiv. Lecturer in universities) at CSIRO Data61, Australia
- Dedicated to blockchain related-technologies, covering consensus, security, privacy, and economics
- Strong publications in terms of productivity and quality
- Balanced research experience and engineering skills
- A good teamwork player, evidenced with the following co-authorships and collaborative projects

EDUCATIONAL BACKGROUND

Ph.D: Swinburne University of Technology & CSIRO Data61, Melbourne, Australia. *Completed 2022*

Superviosr: *Prof. Xiang Yang* (Swinburne), *Dr. Shiping Chen* (CSIRO)

M.Eng: Beihang University (BUAA), Beijing, China. *Completed 2018*

Supervisor: *Prof. Qianhong Wu (*BUAA)

B.Eng: Northwestern Polytechnical University (NPU), Xi'an, China. *Completed 2015*

Major: Electronic Information Science and Technology

RESEARCH

Full publication list can be found at [Google Scholar], [DBLP] and [ResearchGate].

Selected 5 Conference Papers

[5] Xinrui Zhang, Rujia Li, **Qin Wang**, Qi Wang, Sisi Duan. Time-manipulation Attack: Breaking Fairness against Proof of Authority Aura.'' Proceedings of the ACM Web Conference (WWW), 2023. (Core Ranking A*)

[4] **Qin Wang**, Rujia Li, Qi Wang, Shiping Chen, Yang Xiang. ''Exploring Unfairness on Proof of Authority: Order Manipulation Attacks and Remedies.'' ACM Asia Conference on Computer and Communications Security (AsiaCCS), 2022. (Core Ranking A)

[3] Rujia Li, **Qin Wang**, Qi Wang, David Galindo, Mark Ryan. ''SoK: TEE-assisted Confidential Smart Contract Systems.'' 22nd Privacy Enhancing Technologies Symposium (PETS), 2022. (Core Ranking A)

[2] **Qin Wang**, Rujia Li. ''A Weak Consensus Algorithm and Its Application to High-Performance Blockchain.'' IEEE International Conference on Computer Communications (INFOCOM), 2021. (Core Ranking A*)

[1] **Qin Wang**, Jiangshan Yu, Zhiniang Peng, Van Cuong Bui, et al. ''Security Analysis on dBFT protocol of NEO.'' Financial Cryptography and Data Security (FC), pp.20-3. 2020. (Core Ranking A)

Selected 5 Journal Papers

[5] Qin Wang, **Qin Wang**, Shiping Chen, Yang Xiang. ''Exploring Blockchain Interoperability: A Systematic Survey.'' ACM Computing Surveys (CSUR). 2023 (Core Ranking A*)

[4] **Qin Wang**, Jiangshan Yu, Shiping Chen, Yang Xiang. ''SoK: DAG-based Blockchain Systems.'' ACM Computing Surveys (CSUR). 2022 (Core Ranking A*)

[3] **Qin Wang**, Shiping Chen, Yang Xiang. ''Anonymous Blockchain-based System for Consortium.'' ACM Transactions on Management Information Systems (TMIS), 12(3), No.26, pp.1 -- 25. 2021. (Core Ranking A)

[2] **Qin Wang**, Longxia Huang, Shiping Chen, Yang Xiang. ''Blockchain enables Your Bill Safer.'' IEEE Internet of Things Journal (IoTJ), 2020. (Core Ranking A*)

[1] **Qin Wang**, Bo Qin, Jiankun Hu. ''Preserving Transaction Privacy in Bitcoin''. Future Generation Computer System (FGCS), 107, pp.793-804. 2020. (Core Ranking A)

Annual Tech Reports

[3] **Qin Wang***, Rujia Li*, Qi Wang, Shiping Chen, Mark Ryan, Thomas Hardjono. ''Exploring Web3 From the View of Blockchain.'' 2022

[2] **Qin Wang***, Rujia Li*, Qi Wang, Shiping Chen. ''Non-Fungible Token (NFT): Overview, Evaluation, Opportunities and Challenges.'' 2021. (330+ citations)

[1] Xiaotao Feng, **Qin Wang,** Xiaogang Zhu, Sheng Wen. ''Bug Searching in Smart Contract.'' 2019.

International Standards

[2] Shiping Chen, Thomas Hardjono, **Qin Wang**. ''Gateway Identification and Discovery. '' Internet Engineering Task Force (IETF, on Draft stage), 2022.

[1] Saber Yu, **Qin Wang**, Shange Fu, Shiping Chen, Sherry Xu, Jiangshan Yu. ''EIP-5521: A Referable NFT.''Ethereum Improvement Proposals (EIP, on Draft stage), 2021. [doc]

Patent

[2] **Qin Wang**, Qinghua Li, Rujia Li, Xiaoming Wang. "A Method for Distributed Network Authentication based on Hareware Chips". State Intellectual Property Office of the P.R.C (SIPO). Patent Number CN2020106927967. Status: Granted. 2020 (China)

[1] Qianhong Wu, **Qin Wang.** "A Method to Improving Transaction Privacy in Bitcoin". State Intellectual Property Office of the P.R.C (SIPO). Patent Number CN106911470B. Status: Granted. 2017 (China)

WORKING EXPERIENCE

Blockchain DATA Foundation (Full Time), Beijing, China. 2018.04-2018.07 *Position*: R&D of Blockchain.
High Performance Blockchain (Internship), Beijing, China. 2017.10-2018.04 *Position*: R&D of Blockchain.
Samsung China Semicondutor (Internship), Xi'an, China. 2015.02 *Position*: Semiconductor Engineer.
Huawei Authorized Network Training Center (Internship), Xi'an, China. 2014.08 *Position*: Network Engineer.

ACADEMIC SERVICES

Topic Editors: Frontier in Blockchain (DAO) (2023)
Editorial Review Board: IJBASC (2022), Frontier in Blockchain (BC Industry)(2023)
PC member: BSCI@AsiaCCS (2022), BSCI@AsiaCCS (2023)
Invited reviewer: [Publons]
2022: TDSC, TIFS, TPDS, TITS, TNET, TNSE, TNSM, ACM DLT, IoTJ, ESORICS, FGCS, ETT, BSEW, CSI, Cybernetics.
2021: TDSC, ETT, TIFS, PeerJ, ISJ, BCRA.
2020: TDSC, IoTJ, ESORICS, BSCI, IEEE Access.
2019: TDSC, IEEE Access, TII.
External reviewer: ICBC (2021, 2020, 2019), ESORICS (2020), CUSR (2020, 2019), SciSec (2019)

HONORS & AWARDS

Honors

Outstanding Student Cadre, BUAA. 2016
Excellent Student, NPU (Top 30%). 2013

Scholarship

Chinese Government Award for Outstanding PhD Scholar Abroad, (Top 1%). 2021
CSIRO Top-up PhD Scholarship, Australia. 2018-2022
National Scholarship Award, the Ministry of Education of China (Top 1%). 2017
Second Prize of Scholarship, BUAA (Top 30%). 2015
Scholarship of International Class, NPU (Top 20%). 2011-2015

Updated Feb 2023